

Sampling Methods; Exploratory Data Analysis (EDA): Single Variables

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MATH 240

Lecture #4 — Ch. 2.1-2.3, 4.1, 4.4, 4.5, 5.2-5.6

Reminders

- Homework #1 is due **Friday** at 11:59pm
 - ▶ Includes textbook problems and R problem
- Checkpoint #1 is on **Friday** towards the end of class
- Office hours: today from 3-4:30pm in Rocky 106A

Learning Objectives

- **Identify** various components of experimental design in studies
- **Load in** and **work** with data in a .csv format in RStudio
- **Analyze** metrics of a single categorical variable using frequency tables
- **Visualize** a single categorical variable using bar plots

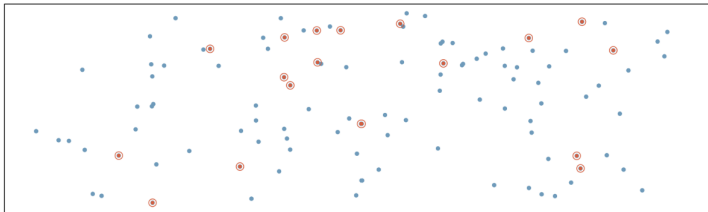
Four Sampling Methods

- ① Simple random sampling (SRS)
 - ▶ Most intuitive; equivalent to picking names out of a hat
- ② Stratified sampling
 - ▶ divide-and-conquer; divide similar cases/makeup (usually with another variable, run SRS)
- ③ Cluster sampling
 - ▶ Break population into heterogeneous clusters; sample all observations from chosen clusters
- ④ Multistage sampling
 - ▶ Break population into clusters randomly; sample randomly within each selected cluster

Examples of the Four Sampling Methods

Let's return to the coffee research problem mentioned at the beginning. Recall that the target population is everyone in the Vassar community.

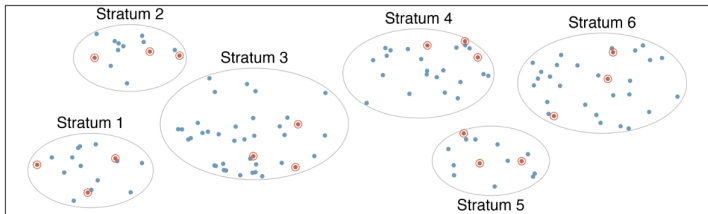
- 1 **Simple random sampling (SRS)**: with a list of all community members' names, we pick 200 names randomly from the list



Examples of the Four Sampling Methods

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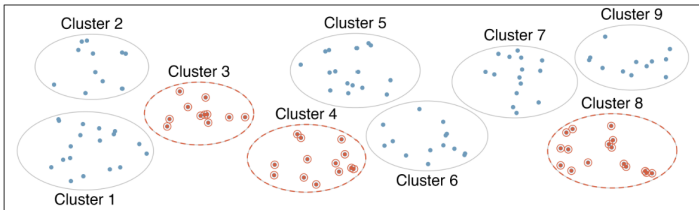
- 2 **Stratified sampling:** Isolating members into strata representing their status at the college (e.g. students, faculty, employees, community members), we pick 50 members randomly from each of the strata



Examples of the Four Sampling Methods

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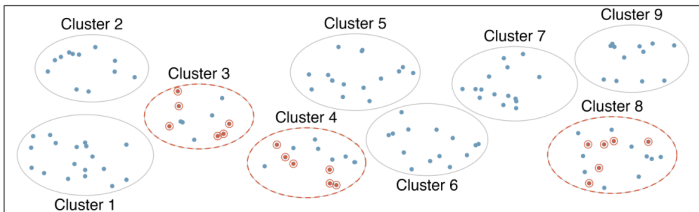
- 3 **Cluster sampling:** break members randomly into many groups of 40, pick 5 clusters and select all members in those chosen clusters



Examples of the Four Sampling Methods

Let's return to the coffee research problem mentioned at the beginning. Recall that the target population is everyone in the Vassar community.

- 4 **Multistage sampling:** break members randomly into many groups of 40 students, pick 10 clusters and select 20 members randomly in those chosen clusters



Experiments vs. Observational Studies

- We also care about a hypothesized relationship between variables, where we have:
 - ▶ A **response variable**: measured outcome of interest
 - ★ e.g. dependent variable, y
 - ▶ **Explanatory variable(s)**: variable potentially explaining or predicting changes in responses
 - ★ e.g. independent variable(s), predictor(s), covariate(s), x
- There are four principles of experimental design

1. Controlling

- Controlling **other variables and factors** that may impact the experiments
- e.g. giving a 12 ounce glass of water to each patient when swallowing a pill, regardless of treatment status

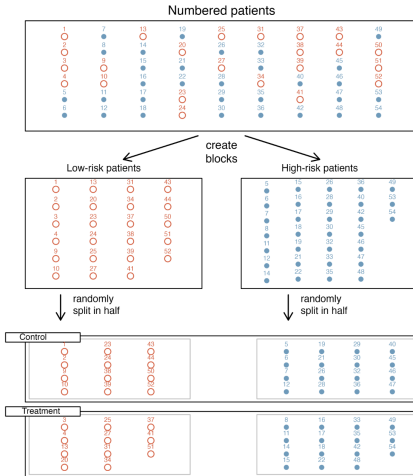
2. Randomization

- Randomization of patients into treatment groups to account for **uncontrolled variables and factors**
 - ▶ Reducing affects of confounding variables (associated with both explanatory and response variables)
 - ▶ e.g. relationship between ice cream sales, hot weather, and drownings
- e.g. dietary habits among patients; need to be randomized into treatment groups

3. Replication

- Need to collect **sufficiently large** sample
- Want multiple subjects (experimental units) per group
- Could replicate an entire study to verify earlier finding

4. Blocking



- Might be variables that **influence** response
- Solution: group individuals based on influential variable into blocks, then randomize cases within each block to treatment groups
- e.g. split patients into low-risk and high-risk blocks, then assign half of the patients in each block to the control and treatment groups respectively

Activity: A Random Experiment

Suppose Vassar wants to test a new food item to start serving at Gordon Commons. They want to sample approximately 200 students from the Vassar student population and identify if a majority of participants liked the new food item, using a proportion to determine this metric.

Consider the following in groups of 2-3:

- What is the population of interest here?
- What would we need to consider to have a random, representative sample? How would we do this?
- How could we implement the following principles in our experiment:
 - ▶ Controlling
 - ▶ Randomization
 - ▶ Replication
 - ▶ Blocking

Working with Data: IMDB Movie Dataset

Today, we'll work with data on movies as a way for us to explore various types of data.

- 3,039 movies made in U.S. between 1929 and 2016, scraped from IMBD (by Dr. Robin Belton)

	movie_title	title_year	budget	log_budget	gross	log_gross
1	Avatar	2009	237000000	19.28357	760505847	20.44949
2	Pirates of the Caribbean: At World's End	2007	300000000	19.51929	309404152	19.55016
3	The Dark Knight Rises	2012	250000000	19.33697	448130642	19.92060
4	John Carter	2012	263700000	19.39032	73058679	18.10677
5	Spider-Man 3	2007	258000000	19.36847	336530303	19.63420
6	Tangled	2010	260000000	19.37619	200807262	19.11786

Figure: First Six Movies in Movies Dataset

Loading in Data

- Open two files from my website
 - ▶ “movies.csv” (dataset we’ll be using)
 - ▶ ‘lect4_eda_temp.qmd” (code we’ll be using)
- Use “read.csv()” command to load dataset in

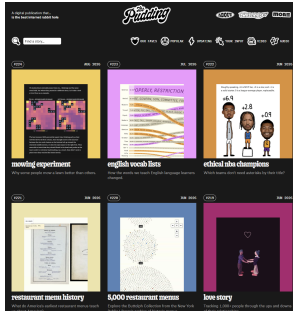
Functions in R

- `dim(data)`
- `nrow(data)`
- `ncol(data)`
- `length(data$column)`
- `typeof(data$column)`

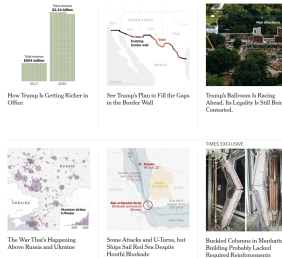
Exploratory Data Analysis (EDA)

- Practice of developing intuition and understanding the characteristics of the existing data, usually through
 - ▶ Data visualizations
 - ▶ Summary tables and metrics
- Helps us inform our understanding of our research question

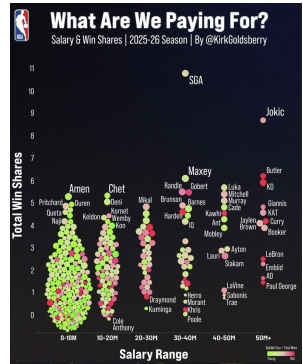
EDA Examples



The Pudding



NYT



Kirk Goldsberry (via IG)

A Summary of Tools

Today, we'll walk through tools to use on **single** variables, such as:

- Frequency and relative frequency tables
- Five-number summaries
- Bar plots
- Histograms
- Density plots
- Box plots

One Categorical Variable: Frequency Tables

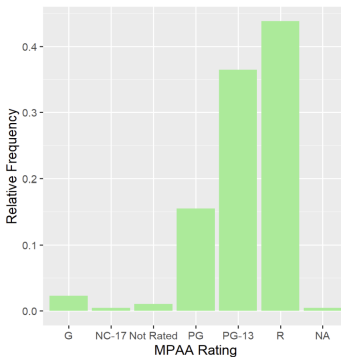
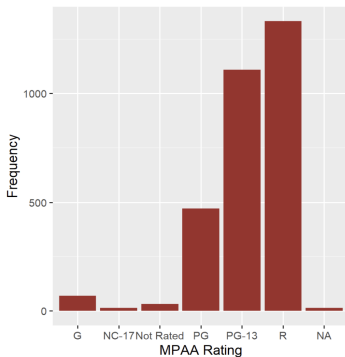
- Tool to get counts of each category in a variable

G	NC-17	Not Rated	PG	PG-13	R
70	13	31	471	1109	1332

Question: How can we get the proportions for each category?

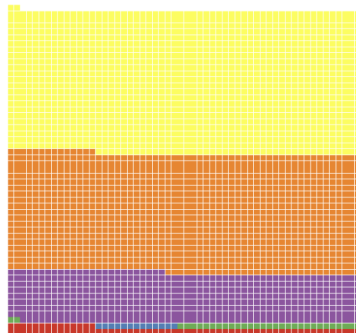
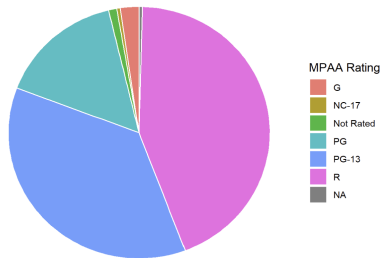
One Categorical Variable: Bar Plot

- Visualizes *counts* of a **categorical** variable
- Traits to note:
 - ▶ Number of categories
 - ▶ Most and least appeared groups
 - ▶ Distribution of counts



One Categorical Variable: Pie and Waffle Charts

- Use of continuous area to show relative size of categories
 - ▶ e.g. pie charts, stacked bar plot
- On a discrete level: waffle chart



The Factor Function

- `factor()` function is useful for associating categories/levels with a variable

```
factor(movies$content_rating, ordered = F,  
       levels = unique(movies$content_rating))
```


For Next Week!

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